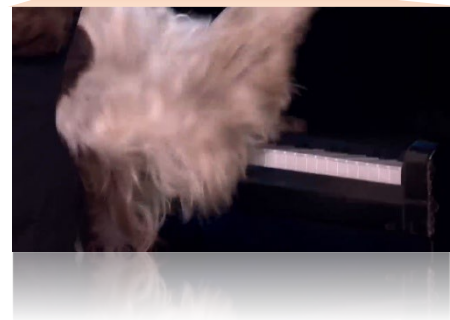
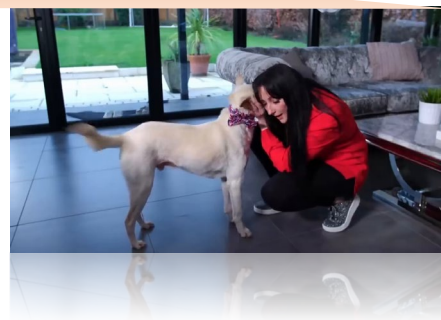
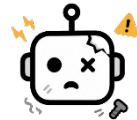


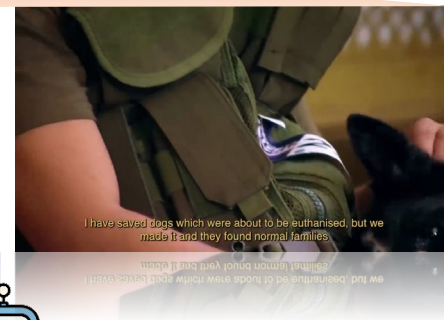
Extensive Raw Video Frames



Uniform Sampling Direct QA



Structured-memory Temporal Re-observation Agent



Question-relevant evidence interval

00:04:45 - 00:05:49

Uniform Sampled 32 Frames



...



Uniform Sampled 64 Frames



...



Qwen3-VL



Redundant frames

Missed short-lived evidence

No reusable memory



One-shot QA



Wrong Answer

Stage 1: Offline Video-level Memory Construction

Segment 0



00:00 - 01:00

Segment 1



01:00 - 02:00

Segment 2



02:00 - 03:00

Segment n



01:42:00 - 01:42:22

Persistent Structured Segment Memory

Scene: Talent-show judging
Actions: Judge speaking
visible text: Talent-show logo
Objects: Judge desk, screen
Keywords: judge, stage, evaluation

...

Scene: Judges and audience area
Actions: reacting to a performance
visible text: Broadcast overlay
Objects: Audience seats, judge desk
Keywords: audience, judges, reaction

...

Scene: Dog performance on stage
Actions: Performer guiding a dog trick
visible text: PRO-TV / talent-show logo
Objects: Dog, performer, stage props
Keywords: dog, performance, trick

Stage 2: Online Query-time Reasoning

Question + Options

What is the **name** of the **dog** that **plays the piano**?
A. Bonnie B. Hurricane
C. Buddy D. Nymeria

Question-aware
Memory Retrieval

Seed Segments

S5 S4 S75

Neighbor Expansion

Primary
evidence

S3 S4

S5

S6 S7

Similar
distractor

S73 S74

S75

S76 S77

Merged Temporal Window

(Remove Overlap)



Local Temporal Re-observation Tool



Qwen3-VL



Visual-grounded
Final QA



Correct Answer

